

DW9714

- 10bit Resolution VCM driver IC with I2C interface

Preliminary
Ver. 0.4
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1. General Description

The DW9714 is single 10-bit DAC with 120mA output current sink capability. Designed for linear control of voice coil motors, the DW9714 is capable of operating voltage to 3.6V. The DAC is controlled via a I²C serial interface that operates DAC by clock rates up to 400kHz.

The DW9714 incorporates with a power-on reset circuit, power-down function, and exactly matched sense resistor. Power-on reset circuit ensure when supply power up, DAC output is to 0V until valid write-bit value takes place. It has a power down features that reduces the current consumption of the device to 1uA maximum.

The DW9714 is designed for auto focus and optical zoom camera phones, digital still cameras, and camcorders applications. The I²C address for the DW9714 is 0x18.

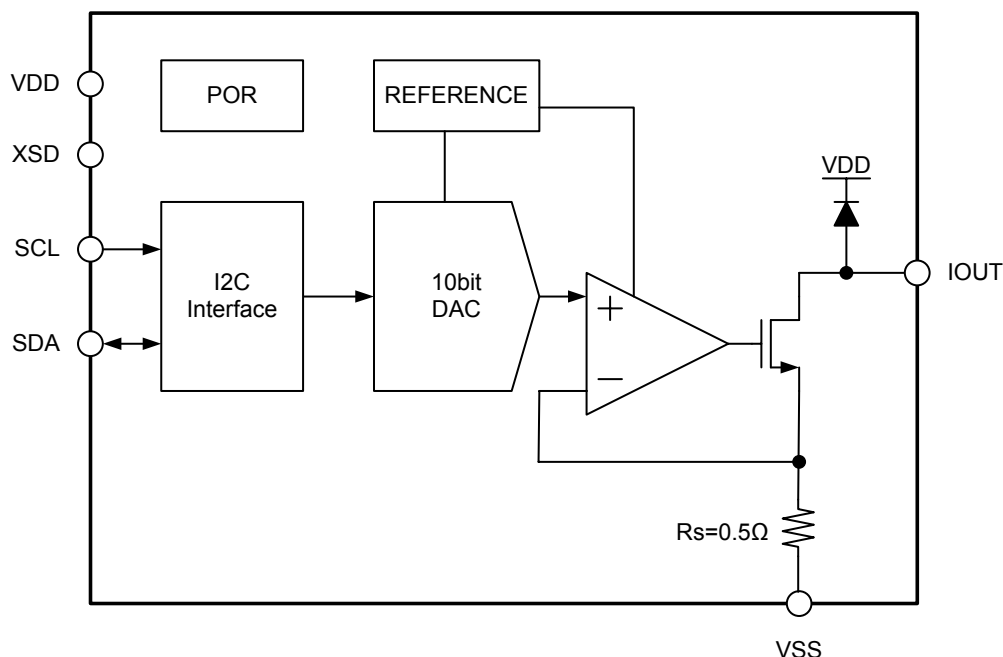
■ Features

- VCM driver for auto-focus
- 10bit resolution current sinking of 120mA for VCM
- VCM slew rate control (SRC) – Linear slope control, Dual level control
- Supply voltage range (VDD) : 2.3V to 3.6V
- Fast mode I2C interface (1.8V interface available)
- Power on reset (POR)
- Package : 0.80mm(W) X 1.20mm(H) X 0.3mm(T) 6pins WLCSP

■ Applications

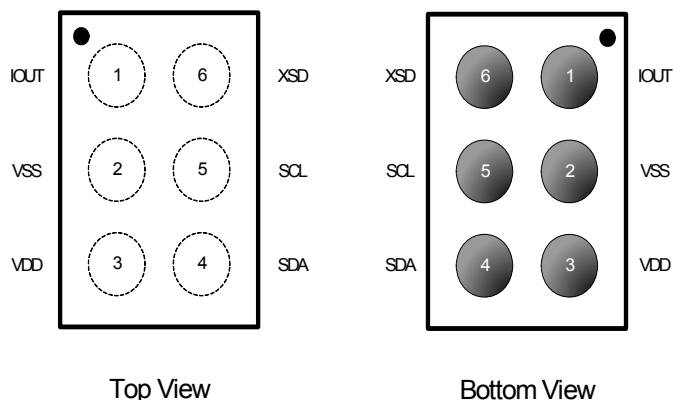
- Digital camera
- Cell phone
- Lens auto focus
- Web camera

2. Block Diagram



3. Pin Information

■ Pin placement and IC dimension



IC size : 0.80(W) X 1.20(L) X 0.30(T) (mm)
Minimum pin pitch = 0.4mm (6pins WLCSP)

■ Pin Description

No.	Pin Name	I/O	Description	Note
1	IOUT	O	Output current sink	
2	VSS	-	Ground	
3	VDD	-	Power supply	
4	SDA	I/O	I2C interface input (DATA)	
5	SCL	I	I2C interface input/output (CLOCK)	
6	XSD ⁽¹⁾	I	Shutdown mode (active low)	

(1) XSD : Shutdown mode (active low)
1: Normal operation mode
0: Shutdown mode

4. Absolute Maximum ratings

Symbol	Parameter	Min.	Max.	Unit
VDD	Power supply voltage	-0.3	4.5	V
Vin	Control input voltage	-0.3	VDD+0.3	V
Vhbm	Human body model		2	KV
Vmm	Machine model		200	V
Topr	Operating temperature range	-35	85	℃
Tj	Junction temperature		150	℃

Note> Continuous Power Dissipation (Ta=25℃)
0.80mm X 1.20mm WLCSP, 100 ℃/W

5. Recommended Operating condition

Symbol	Parameter	Min.	Typ.	Max.	Unit
VDD	Power supply voltage	2.3	2.8	3.6	V
Vin	Control input voltage	1.8	2.8	VDD	V
SCL	I2C bus transmission rate			400	kHz

■ Absolute maximum ratings

Use of the IC in excess of absolute maximum ratings such as the applied voltage or operating temperature range (Topr) may result in IC damage. Assumptions should not be made regarding the state of the IC (short mode or open mode) when such damage is suffered. The implementation of a physical safety measure such as a fuse should be considered when use of the IC in a special mode when the absolute maximum ratings may be exceeded is anticipated.

6. Electrical Specification

(VDD=2.3 to 3.6V, Vin=1.8V to VDD, Ta= -35 to 85 °C, unless otherwise specified. Typical values are at 25 °C)

Characteristic	Symbol	Condition	Min.	Typ.	Max.	Unit
Overall						
Supply Voltage	V _{DD}		2.3		3.6	V
V _{DD} Current	I _{SD}	Shutdown mode	-1		+1	uA
	I _{PD}	Power down mode	-1		+1	uA
	I _Q	Quiescent mode	0.24	-	0.35	mA
Logic input / output (XSD)						
Input current			-1		+1	uA
Low Level Input Voltage	V _{IL}				0.54	V
High Level Input Voltage	V _{IH}		1.26			V
Logic input / output (SCL, SDA)						
Input current			-1		+1	uA
Low Level Input Voltage	V _{IL}				0.54	V
High Level Input Voltage	V _{IH}		1.26			V
Glitch rejection				50		ns
VCM driver						
Current resolution		117.3uA/LSB		10		bits
INL	INL		-4		+4	LSB
DNL	DNL		-1		+1	LSB
Zero code error	ZCE	Zero data loaded to DAC	-1		+1	mA
IOUT compliance voltage ⁽¹⁾		Output current = 100mA	150			mV
Maximum output current	I _{max}			120 ⁽³⁾		mA
Power on time ⁽²⁾	TPON			12		ms

(1) The output compliance voltage is guaranteed by design and characterization, not mass production test.

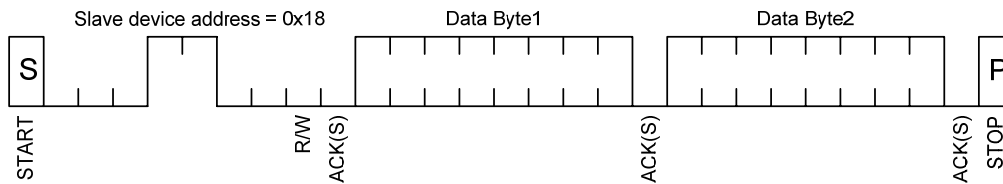
(2) DW9714 requires waiting time of 12ms after power on. During this waiting time, the offset calibration of internal amplifier is operating for minimization of output offset current .

(3) Maximum output current can be set 60mA to 140mA.

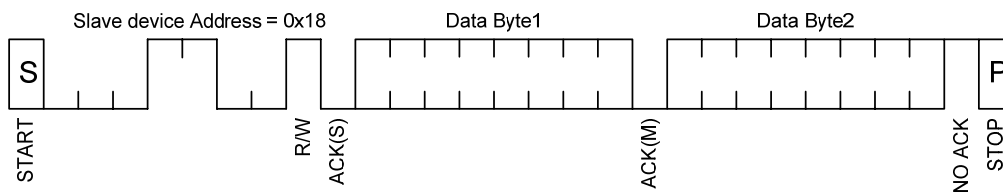
7. Register

7.1. I2C format

■ Write Operation



■ Read Operation



7.2. Register Format

Byte1								Byte2							
PD	FL AG	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0	S3	S2	S1	S0

PD : Power down mode

1: Power down mode (active high)

0: Normal operation mode

FLAG : FLAG must keep “L” at writing operation.

D[9:0] : Data input

Output current = (D[9:0]/1023) X 120mA

Max current = 120mA +/- 5%

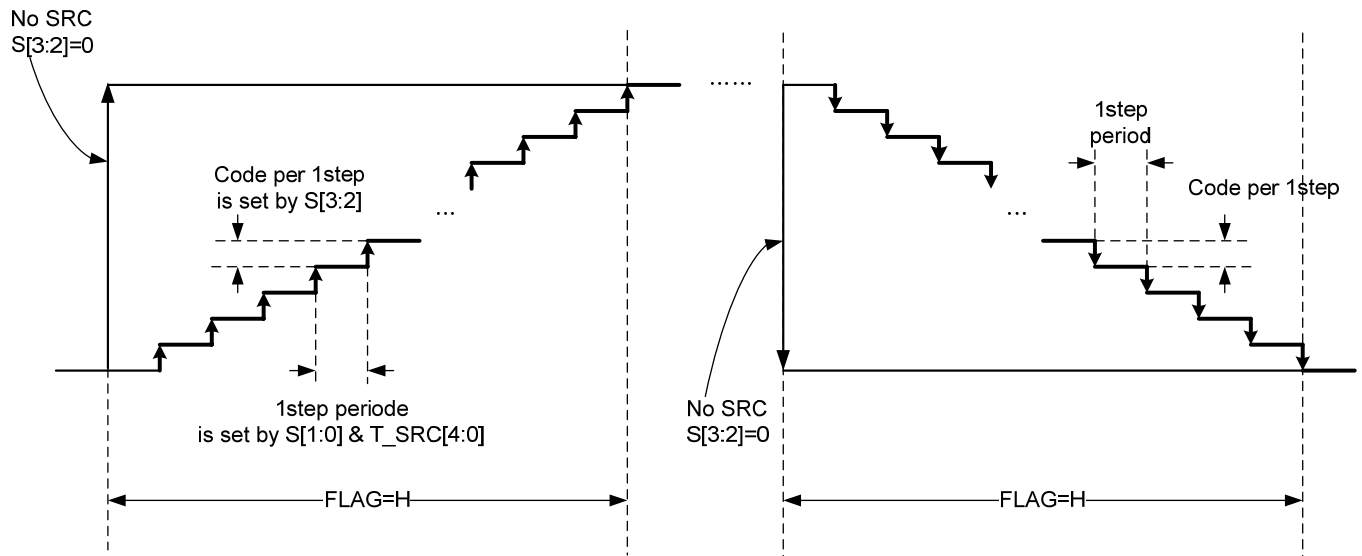
S[3:2] : Codes per step

S[3:2]	Codes per step
0	0 (no SRC) – direct driving
1	1
2	2
3	4

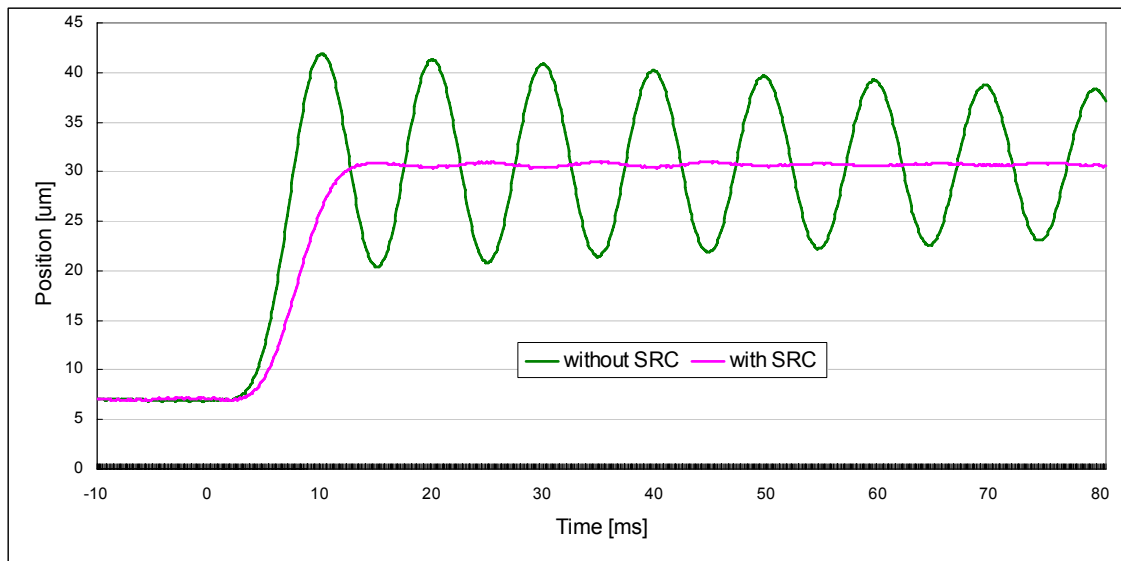
S[1:0] : Step period

SRCT[1:0]	Period [us]
0	81
1	162
2	324
3	648

■ SRC control scheme

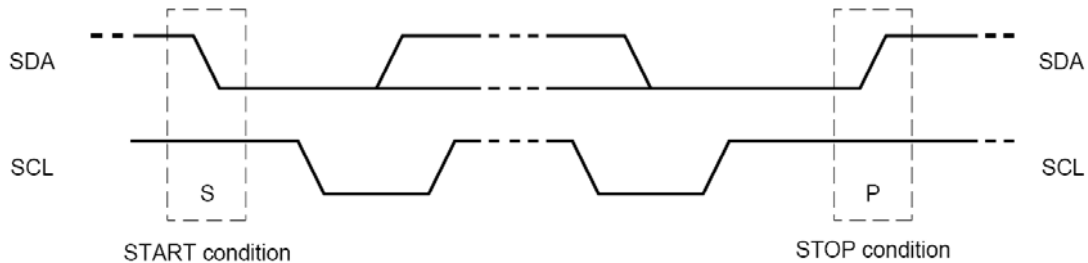


7.3. SRC Test Results



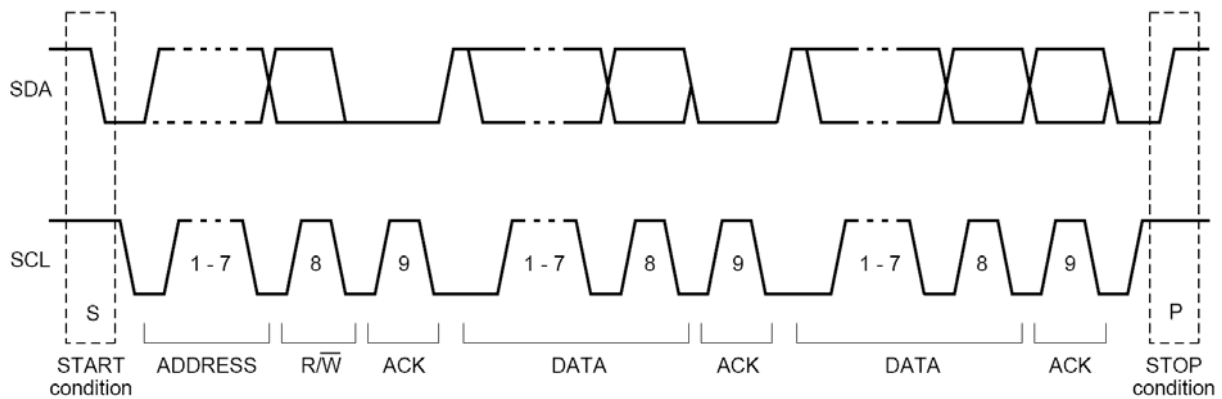
8. I2C Protocol

■ Start and Stop condition



Within the procedure of the I2C-bus, unique situations arise which are defined as START (S) and STOP (P) conditions. A HIGH to LOW transition on the SDA line while SCL is HIGH is one such unique case. This situation indicates a START condition. A LOW to HIGH transition on the SDA line while SCL is HIGH defines a STOP condition.

■ Complete I2C Data Transfer



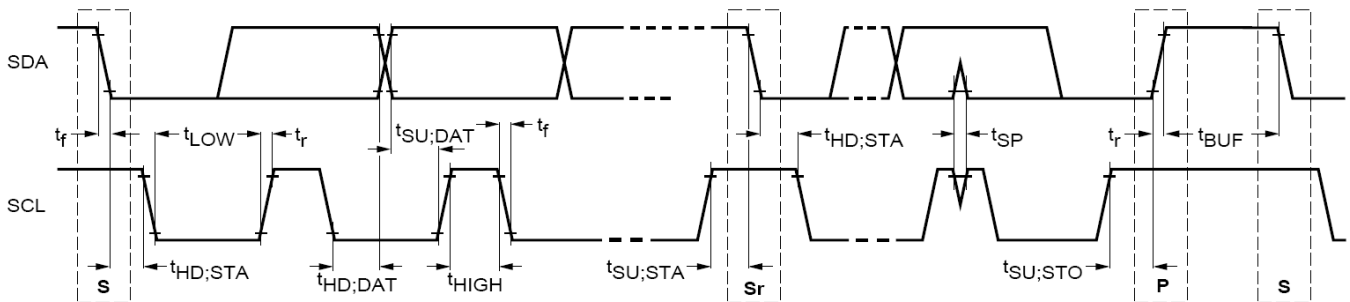
Data transfers follow the format. After the START condition (S), a slave address is sent. A data transfer is always terminated by a STOP condition (P) generated by the master. However, if a master still wishes to communicate on the bus, it can generate a repeated

■ I2C timing

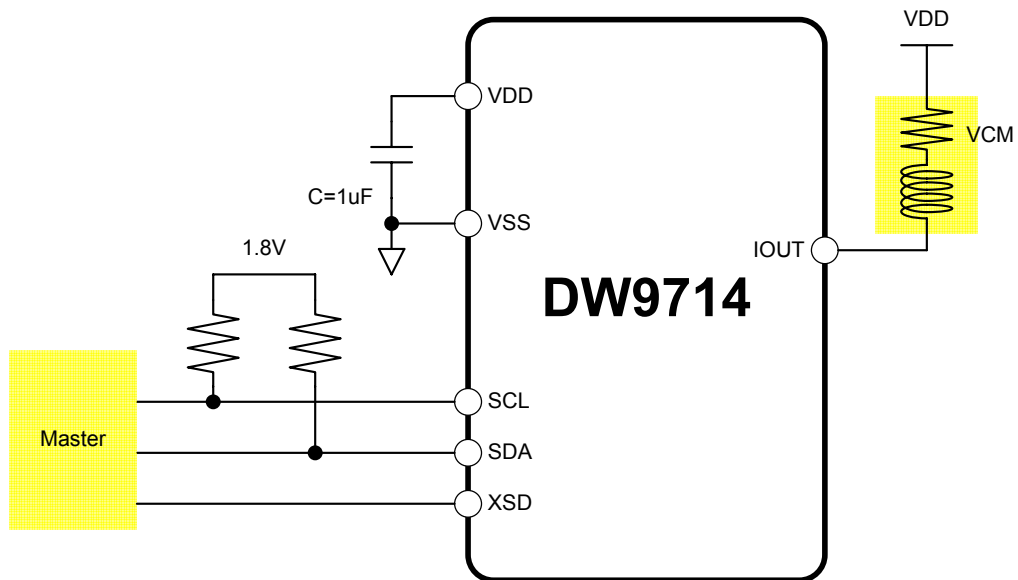
Parameter	Symbol	Min.	Max.	Unit
SCL clock frequency	f _{SCL}	0	400	kHz
Hold time (repeated) START condition.	t _{HD;STA}	0.6	-	us
Low period of the SCL clock	t _{LOW}	1.3	-	us
High period of the SCL clock	t _{HIGH}	0.6	-	us
Set-up time for a repeated START condition	t _{SU;STA}	0.6	-	us
Data hold time	t _{HD;DAT} ⁽¹⁾	-	0.9	us
Data set-up time	t _{SU;DAT}	100	-	ns
Rise time of both SDA and SCL signals	t _r	20+0.1C _b ⁽²⁾	300	ns
Fall time of both SDA and SCL signals	t _f	20+0.1C _b ⁽²⁾	300	ns
Set-up time for STOP condition	t _{SU;STO}	0.6	-	us
Bus free time between a STOP and START condition	t _{BUF}	1.3	-	us
Capacitive load for each bus line	C _b	-	400	pF
Pulse width of spike suppress	t _{SP}	0	50	ns

(1) A master device must provide a hold time of at least 100ns for the SDA signal to bridge the undefined region of the falling edge of SCL. The maximum t_{HD;DAT} has only to be met if the device does not stretch the LOW period (t_{LOW}) of the SCL signal.

(2) C_b is the total capacitance of one bus line in pF, t_r and t_f are measured between 0.3V_{DD} to 0.7V_{DD}.

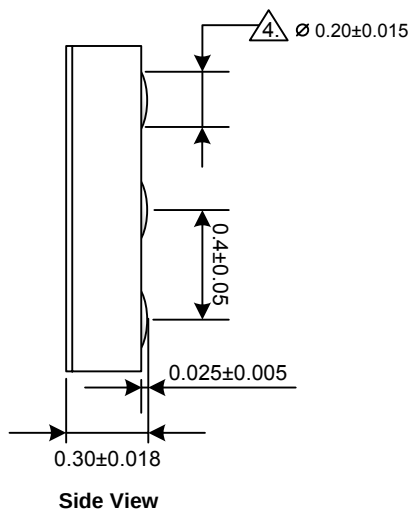
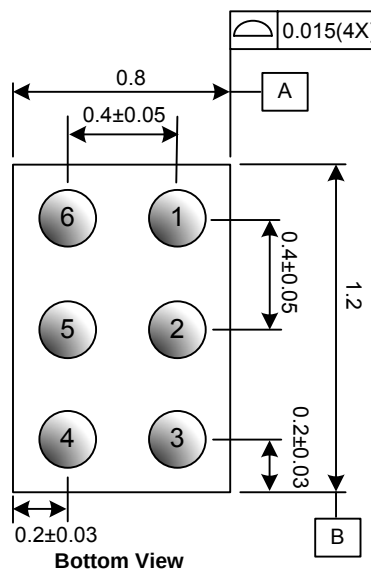
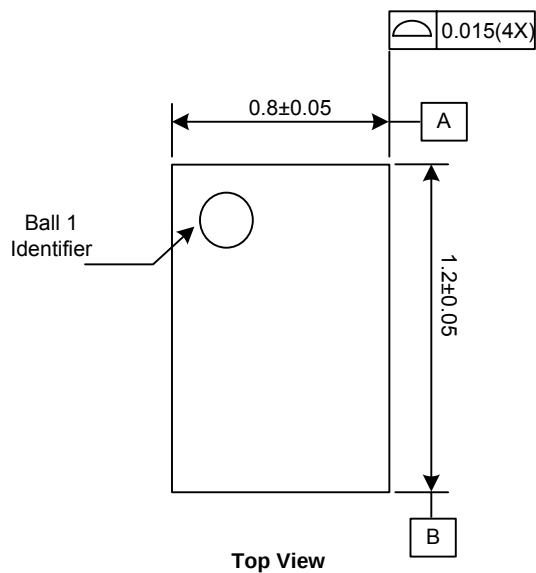


9. Typical Application Circuit



10. Package Dimension (6 WLCSP 0.80 x 1.20 x 0.30)

* Unit : mm



NO	NAME	I/O
1	IOOUT	O
2	VSS	-
3	VDD	-
4	SDA	I/O
5	SCL	I
6	XSD	I